Lecture 6: Dependent Sources and the Ideal Transformer

EE201 – Circuit Theory I

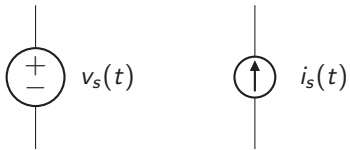
M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

current voltage power magnetic flux

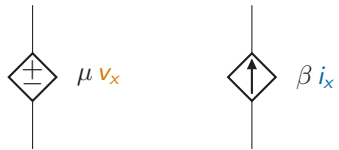
Sources controlled from elsewhere

independent



decides
its own value

dependent



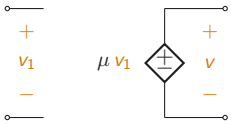
copies a variable
from elsewhere, scaled

Dependent (controlled) source

A voltage or current source whose value is a function of a voltage or current in *another* branch — the controlling variable. Almost always the function is a constant times one variable: a linear dependent source.

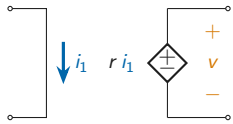
The four kinds

1. VCVS — voltage-controlled voltage source



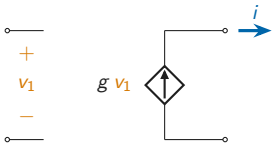
$v = \mu\, v_1$

2. CCVS — current-controlled voltage source



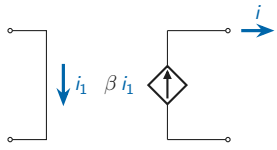
$v = r\, i_1$

3. VCCS — voltage-controlled current source



$i = g\, v_1$

4. CCCS — current-controlled current source



$i = \beta\, i_1$

The name reads in order: what it senses, then what it produces.

The constant, and its unit

Kind	Equation	The constant	Unit
VCVS	$v = \mu\, v_1$	voltage gain	none (V/V)
CCVS	$v = r\, i_1$	transresistance	ohm, Ω (V/A)
VCCS	$i = g\, v_1$	transconductance	siemens, S (A/V)
CCCS	$i = \beta\, i_1$	current gain	none (A/A)

The unit is always what comes out over what goes in.

Trans- when the two differ: a transresistance turns a current into a voltage, so it is measured in ohms.

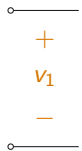
More general forms

Several variables: $\mu_a v_a + \mu_b v_b$ — one source per term.

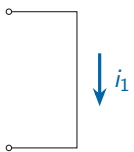
Affine: $\mu v_x + V_0$ — plus an independent source V_0.

Nonlinear: $i = f(v_x)$ — exists, not used in EE201.

The sensing side



senses a voltage:
open circuit, $i = 0$

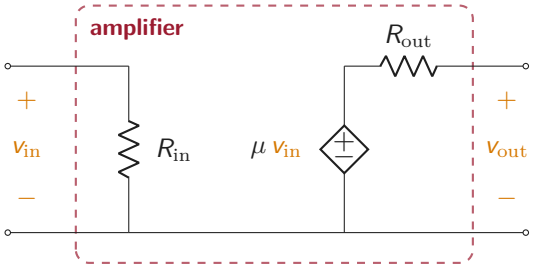


senses a current:
short circuit, $v = 0$

It senses without disturbing anything.

In real diagrams this side is not drawn: the controlling variable is the voltage across, or the current through, an element that is already there — labelled v_x, i_x, i_1.

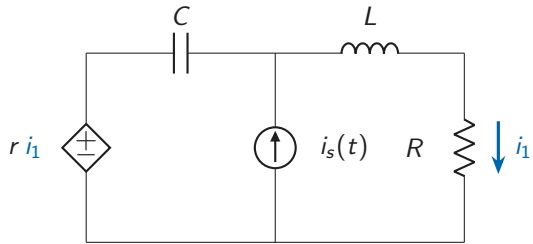
Why we need them



The simplest model of anything that amplifies. Many components and integrated circuits are modelled as dependent sources — and some are designed to be one.

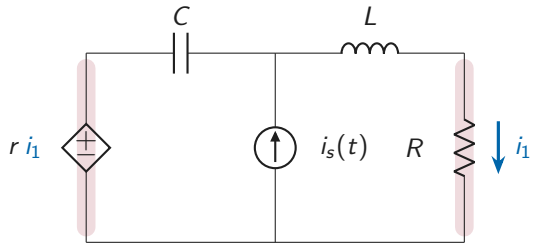
VCVS	op-amp	$\mu \approx 10^5$
CCVS	current-sense amplifier	r (Ω)
VCCS	MOSFET, transconductance amp.	g_m (S)
CCCS	bipolar transistor	$\beta \approx 100$

Example: what the diamond says



Which equation does the dependent source contribute?

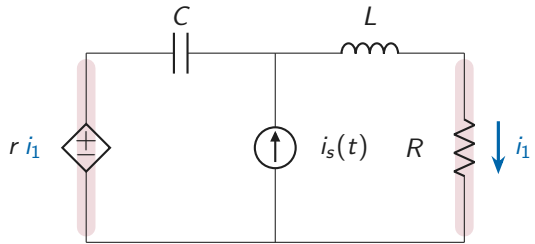
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Which equation does the dependent source contribute?

A CCVS: $v = r\, i_1$ — and i_1 flows through R, at the far right.

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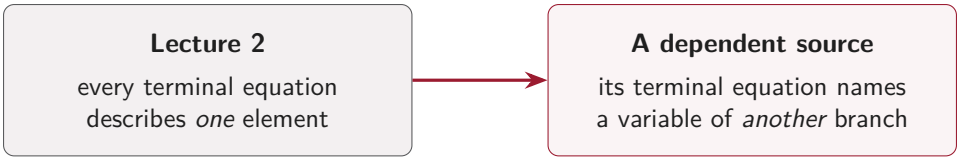


Which equation does the dependent source contribute?

A CCVS: $v = r\, i_1$ — and i_1 flows through R, at the far right.

$n = 4$ nodes and $b = 5$ branches: $2b = 10$ unknowns, $b = 5$ terminal equations.
Four speak about one element each. The fifth links two branches.

What is new — and what is not



KCL and KVL are unchanged. Still $2b$ equations for $2b$ unknowns.

Two meanings of "controlled"

Lecture 3: a *voltage-controlled resistor* has $i = f(v)$ of its own voltage. A voltage-controlled *source* takes its value from another branch.

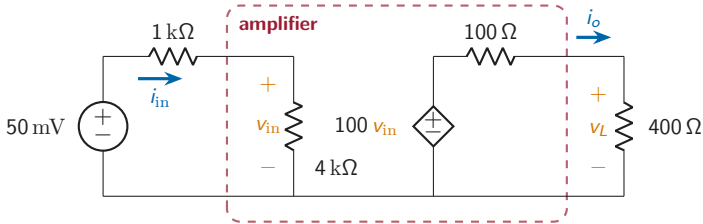
No new method

Recipe

1. Treat the dependent source as if it were independent, with its value written as μv_x, $r i_x$, $g v_x$ or βi_x.
2. Write KCL, KVL and the terminal equations as usual.
3. Add one equation that gives the controlling variable in terms of your unknowns.

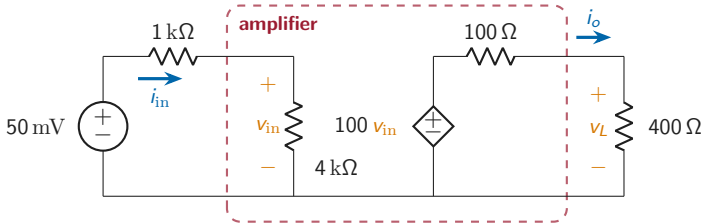
The power check $\sum_k p_k = 0$ still works — include the dependent source.

Example: an amplifier



A microphone drives an amplifier with $\mu = 100$. Find v_L.

Example: an amplifier



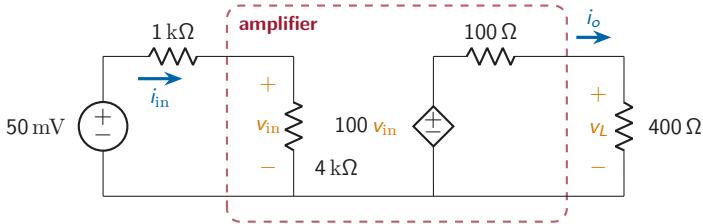
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Input side — a divider:

$$i_{in} = \frac{50\,\text{mV}}{5\,\text{k}\Omega} = 10\,\mu\text{A}$$

$$v_{in} = 4\,\text{k}\Omega \times 10\,\mu\text{A} = 40\,\text{mV}$$

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Output side — the diamond is $4\,\text{V}$:

$$i_o = \frac{4\,\text{V}}{500\,\Omega} = 8\,\text{mA}$$

$$v_L = 3.2\,\text{V}, \qquad \frac{v_L}{v_s} = 64$$

Where the power comes from

input side	
microphone	$-0.5\,\mu\mathrm{W}$
R_s	$+0.1\,\mu\mathrm{W}$
R_{in}	$+0.4\,\mu\mathrm{W}$

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output side	
dependent source	$-32\,\mathrm{mW}$
R_{out}	$+6.4\,\mathrm{mW}$
R_L	$+25.6\,\mathrm{mW}$

Each side sums to zero.

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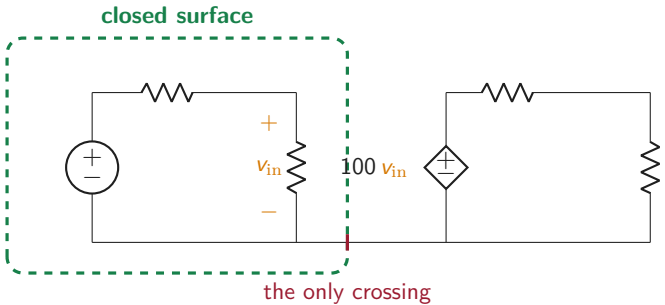
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Each side sums to zero.

The load gets 51 200 times what the microphone supplies.
The dependent source delivered it. A real amplifier takes it from its power supply, which the model does not draw.

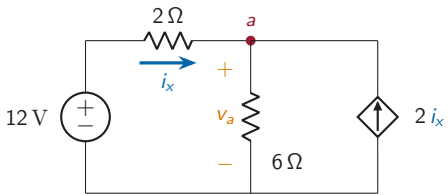
The shared wire carries no current



KCL on the surface (Lecture 2): one branch crosses it, so its current is zero.

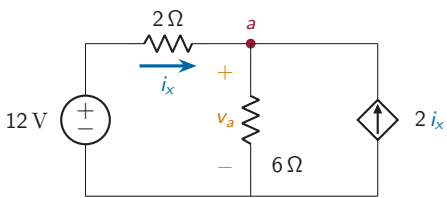
The two sides interact only through v_{in}.

Example: when the controlling variable is unknown



Find i_x and v_a.

Example: when the controlling variable is unknown



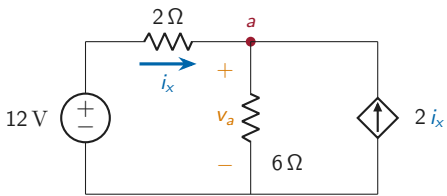
Find i_x and v_a.

Equations — i_x is already an unknown:

$$\text{KVL:}\quad -12 + 2i_x + v_a = 0$$

$$\text{KCL at } a\text{:}\quad -i_x - 2i_x + \tfrac{v_a}{6} = 0 \;\Rightarrow\; v_a = 18\, i_x$$

Example: when the controlling variable is unknown



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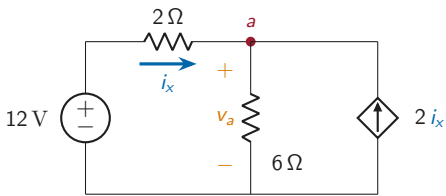
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Solution: $-12 + 20\, i_x = 0$

$$i_x = 0.6\,\text{A}, \qquad v_a = 10.8\,\text{V}$$

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Power check:

$$\underbrace{-7.2}_{12\,\text{V}} + \underbrace{0.72}_{2\Omega} + \underbrace{19.44}_{6\Omega} \underbrace{-\,12.96}_{\text{CCCS}} = 0$$

The CCCS carries $1.2\,\text{A}$ *out of* its + terminal: it delivers.

What a dependent source cannot do

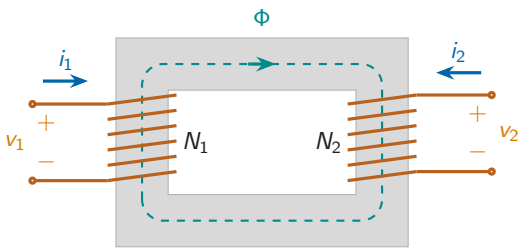
It cannot start anything

Switch the $12\,\mathrm{V}$ off: $2i_x + v_a = 0$ and $v_a = 18i_x$, so $20\,i_x = 0$.
Every voltage and current is zero — the dependent source's too.

It is never switched off

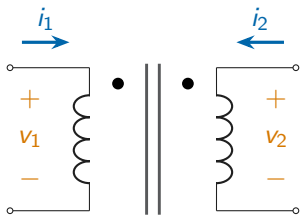
Later we switch sources off one at a time (superposition, Thévenin). Only independent sources are switched off: a dependent source is part of how the circuit behaves, not an input.

Two coils, one core



- **Lecture 1:** no changing magnetic field links the loops.
- **Lecture 2:** if we *want* such a link, draw it as an element.
- **Lecture 5:** a coil's field passes through the coil itself.
- **Today:** two coils on one iron core — the field of each passes through the other.

The ideal transformer



Two equations

$$\frac{v_1}{N_1} = \frac{v_2}{N_2} \qquad N_1 i_1 + N_2 i_2 = 0$$

v positive and i entering at the dots.

Equivalently $v_2 = \frac{N_2}{N_1} v_1$, $i_2 = -\frac{N_1}{N_2} i_1$.

The dots act like reference marks. Move one dot and both equations change sign.

Where the two equations come from

Voltages: one flux through every turn

Each turn sees $d\Phi/dt$. N turns in series: $v = N\, d\Phi/dt$.

$$\frac{v_1}{N_1} = \frac{v_2}{N_2} = \frac{d\Phi}{dt}$$

Currents: an ideal core needs no current

The turn-weighted currents must cancel: $N_1 i_1 + N_2 i_2 = 0$.
A real core needs a small magnetising current; the ideal one needs none.

With k windings on one core: $\dfrac{v_1}{N_1} = \cdots = \dfrac{v_k}{N_k}$ and $N_1 i_1 + \cdots + N_k i_k = 0$.

No derivatives — and no DC

The equations are algebraic: no d/dt, no memory.

The ideal transformer is a resistive element — listed with resistors and dependent sources in the course outline, even though it is built from coils.

A real transformer needs a changing flux

Feed it DC and, once the current settles, the flux is constant: the secondary voltage is zero and the primary is just a coil — a short circuit at DC (Lecture 5). Use the ideal model for changing waveforms only.

The ideal transformer absorbs no power

Each winding obeys the PSC, so $v_j i_j$ is what it absorbs.

$$p(t) = \sum_{j=1}^{k} v_j i_j = 0 \qquad \text{at every instant}$$

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The first equation gave λ; the second made the sum vanish. Both are needed.

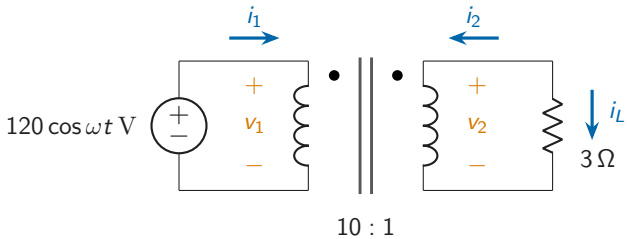
Dissipate, store, or pass through

Element	Absorbed power	The energy is
resistor	$p \geq 0$	dissipated
capacitor, inductor	p can change sign	stored, then returned
ideal transformer	$p = 0$ at every instant	passed straight through

Two windings: $v_1 i_1 = -v_2 i_2$.

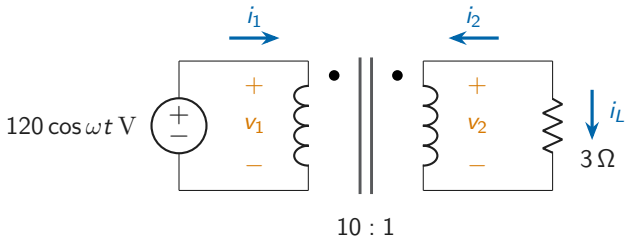
Voltage multiplied by N_2/N_1, current divided by N_2/N_1 — the product unchanged.

Example: a step-down transformer



Find the currents and the power on each side.

Example: a step-down transformer



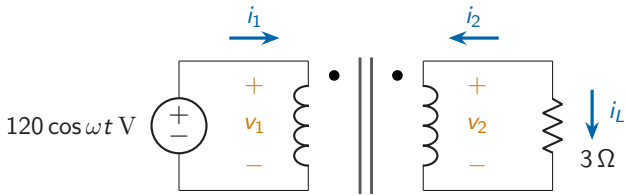
Find the currents and the power on each side.

$v_2 = \frac{1}{10} v_1 = 12\cos\omega t$ V

$i_L = v_2/3 = 4\cos\omega t$ A, so $i_2 = -4\cos\omega t$ A

$i_1 = -\frac{1}{10} i_2 = 0.4\cos\omega t$ A

Example: a step-down transformer



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$p_1 = v_1 i_1 = 48\cos^2\omega t$ W

$p_2 = v_2 i_2 = -48\cos^2\omega t$ W

$p_1 + p_2 = 0$ — at every instant.

Seen from the source

$$\frac{v_1}{i_1} = \frac{120}{0.4} = 300\,\Omega = 10^2 \times 3\,\Omega$$

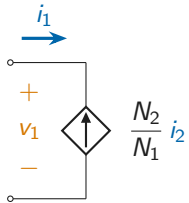
The source cannot tell the transformer and its $3\,\Omega$ load from a single $300\,\Omega$ resistor.

A transformer rescales resistance

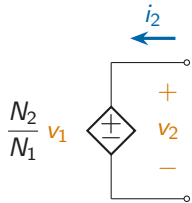
$$\frac{v_1}{i_1} = \left(\frac{N_1}{N_2}\right)^2 R_L$$

That is how an $8\,\Omega$ loudspeaker is made to look like a much larger resistance to the amplifier driving it.

A transformer is two dependent sources

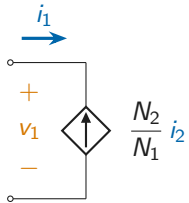


CCCS, controlled by i_2

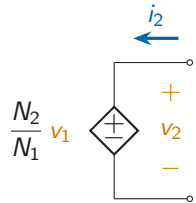


VCVS, controlled by v_1

A transformer is two dependent sources



CCCS, controlled by i_2

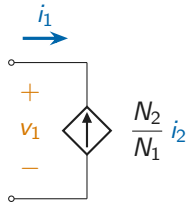


VCVS, controlled by v_1

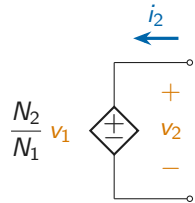
CCCS: $i_1 = -\frac{N_2}{N_1} i_2$ is $N_1 i_1 + N_2 i_2 = 0$.

VCVS: $v_2 = \frac{N_2}{N_1} v_1$.

A transformer is two dependent sources



CCCS, controlled by i_2



VCVS, controlled by v_1

CCCS: $i_1 = -\frac{N_2}{N_1} i_2$ is $N_1 i_1 + N_2 i_2 = 0$. VCVS: $v_2 = \frac{N_2}{N_1} v_1$.

Powers: $-\frac{N_2}{N_1} v_1 i_2$ and $+\frac{N_2}{N_1} v_1 i_2$ — one delivers exactly what the other absorbs.

Summary

- A **dependent source** (diamond) takes its value from a voltage or current in another branch: VCVS, CCVS, VCCS, CCCS. Unit of the constant: out over in. Almost always linear in one variable.
- Its terminal equation names a foreign variable. Write KCL, KVL and terminal equations as usual, plus one equation for the controlling variable.
- It can **deliver power** — that is amplification — but it **cannot start anything**, and it is never switched off.
- **Ideal transformer**: $v_1/N_1 = v_2/N_2$, $N_1 i_1 + N_2 i_2 = 0$, signs set by the dots. Resistive, but a real one needs changing waveforms.
- It **absorbs no power at any instant** and rescales a load by $(N_1/N_2)^2$.

Next lecture

From circuit to graph

Labelling the nodes and branches of a circuit so that its $2b$ equations can be written systematically — the first step towards node and mesh analysis.

Questions?